On the Proof of the Fifth Postulate (the Parallel Axiom) in Euclidean Geometry

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Abstract

The framework of Euclidean geometry lacks a definitive criterion to distinguish straight lines from curves, prompting the introduction of the Fifth Postulate (the Parallel Axiom) as a logical patch. However, this postulate can be proven without new axioms by utilizing the concept of area. The core distinction lies in the fact that equal line segments can coincide after translation on a plane, but not necessarily on curved surfaces. This translational property enables the proof of the Parallel Postulate on a plane without contradicting the existence of non-Euclidean geometries. The fundamental distinction between Euclidean and non Euclidean geometry resides in the first postulate: an Euclidean line is uniquely determined by two points alone, while a non Euclidean line presupposes infinitely many points that fix an underlying surface.

Keywords: Euclidean geometry, the Fifth Postulate, the Parallel Axiom, proof
MSC2020: 51M05

1 Introduction

In plane geometry, the five Euclidean postulates are as follows: [1], pp. 156–157.

Postulate 1: To draw a straight line from any point to any point.
Postulate 2: To produce a finite straight line continuously in a straight line.

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Postulate 3: To describe a circle with any center and any distance.
Postulate 4: That all right angles are equal to one another.
Postulate 5 (Parallel Postulate): Through a given point not on a given straight line, there exists exactly one straight line parallel to the given line.

The five Common Notions are as follows: [1], pp. 157–158.

C.N. 1: Quantities equal to the same quantity are equal to each other.
C.N. 2: If equal quantities are added to equal quantities, the sums are equal.
C.N. 3: If equal quantities are subtracted from equal quantities, the differences are equal.
C.N. 4: Magnitudes that coincide are equal.
C.N. 5: The whole is greater than any of its parts.

2 Notation

Throughout this paper, the following notation is adopted:

Prop. I. n: Proposition n in Book I of Euclid's *Elements*.

Postulate n: The n-th Postulate in Euclid's *Elements*.

C.N. n: The n-th Common Notion in Euclid's *Elements*.

Def. n: The n-th Definition in Euclid's *Elements*.

Corollary n: The n-th corollary derived from a proposition.

m**:** The length of a line segment (e.g., AB = BC = m, where m is finite).

M, L**:** The length of a ray line (M, L are infinite).

q, Q**:** Integers used in the bisection of angles (Q denotes a specific value of q).

β**:** The measure of an angle in degrees ($0 < \beta \leq 180$).

$t_{AB \parallel CD}$**:** The area between two isogonal parallel lines AB and CD.

$t_{\angle BAC}$**:** The area inside $\angle BAC$.

$A * B * C$**:** Points A, B, C are collinear and B lies between A and C.

3 Six Corollaries

Euclid further proves Propositions I.9, I.11, I.12, I.16, I.27 and I.28 without relying on the Fifth Postulate ([1], pp. 267–283). These propositions will be used directly throughout this paper.

3.1 Corollary 1

As shown in Figure 1, point B is on ray AC and lies between A and C. Suppose $AB = m$, the length of ray BC is M (where $M \to \infty$), and the length of ray AC is L (where $L \to \infty$).



Figure 1

By C.N. 5, $L = M + m > M$. By C.N. 4, ray AC may coincide with the ray BC, so that $L = M$. Equality of two quantities X and Y is written as $X = Y$. We have

$$X = Y \iff X - Y = 0 \quad \text{or} \quad Y/X = 1.$$

Which characterization should we adopt? For finite X, Y, the two formulations are equivalent, but they differ when X and Y are infinite quantities because $L - M = M + m - M = m \neq 0$. To eliminate the apparent contradiction between C.N. 4 and C.N. 5 when dealing with infinite quantities, we should adopt the condition $X/Y = 1$. That is,

$$\frac{M}{L} = \frac{M}{M+m} = 1, \quad \text{valid as } M \to \infty.$$

If this choice enables us to prove the Fifth Postulate, then it must be the choice intended by God. That is, two quantities X and Y are equal whenever they can

coincide, namely, $X/Y = 1$, rather than $X - Y = 0$. We have

$$\begin{aligned} X = Y &\iff X/Y = 1 \\ &\iff X/Y - 1 = 0 \\ &\iff \frac{X - Y}{X} = 0 \\ &\iff X - Y = \Delta X, \quad \frac{\Delta X}{X} = 0 \\ &\iff Y = X + \Delta X, \quad \frac{\Delta X}{X} = 0. \end{aligned}$$

3.2 Corollary 2

According to Postulate 2, the straight line AB divides the plane into two half-planes, as shown in Figure 2, referred to as the $\mathcal{H}^-_{AB}$ and the $\mathcal{H}^+_{AB}$. Lines and planes can all be regarded as sets composed of points. $\mathcal{H}^-_{AB} \cap \mathcal{H}^+_{AB} = AB$.

Let line DE pass through point C on line AB, where DE and AB are distinct lines. Suppose C lies between D and E ($D * C * E$). Then rays CD and CE cannot lie in the same half-plane. Otherwise, $\angle DCE < 180°$, which contradicts the fact that DE is a straight line.

If $P \in \mathcal{H}^-_{AB}$, $Q \in \mathcal{H}^+_{AB}$, with neither P nor Q lying on line AB, then line PQ must intersect line AB.

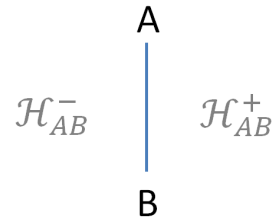


Figure 2

Euclid repeatedly employs the notion of points lying on the same side of a straight line. This paper attempts to prove this corollary.

3.2.1 Same Side and Opposite Sides: Definitions

As shown in Figure 3, let P and Q be two points on a plane not lying on the straight line AB. Draw perpendiculars PC and QD from P and Q to AB, with feet of perpendicularity C and D respectively, where $C \neq D$ (Prop. I. 12 [1], p. 271).

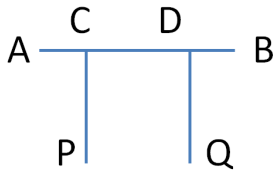


Figure 3

When moving from P to C, then to D, and finally to Q, two successive clockwise rotations of 90° are performed, summing to a total rotation of 180°. In this case, P and Q are said to lie on the same side of AB.

If one 90° rotation is clockwise and the other counterclockwise (opposite directions), yielding a net rotation of 0°, then P and Q are said to lie on opposite sides of AB, as illustrated in Figure 4.

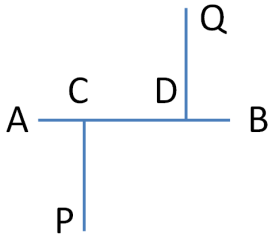


Figure 4

If $C = D$, the total rotation angle is equal to the case where $C \neq D$.

3.2.2 Every angle of a triangle is less than 180 degrees

As shown in Figure 5, if $\angle BAC$ in $\triangle BCA$ is greater than 180°, the extension of side CA will lie inside the triangle. By Postulate 2, a straight line can be extended indefinitely, so this line must exit triangle ABC and stretch to infinity. The line AC already intersects AB and BC. No matter which side it exits through, two straight lines would share two intersection points, which contradicts Postulate 1.

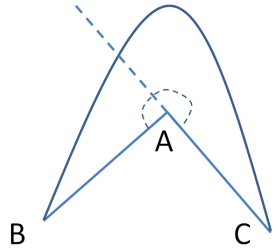


Figure 5

A straight line cannot intersect itself. As illustrated in Figure 6, suppose a straight line intersects itself at point E, forming a closed loop. Choose a point D not equal to E on the loop; then two distinct straight segments connect D and E, contradicting Postulate 1. Therefore, every angle of a triangle is less than 180°.

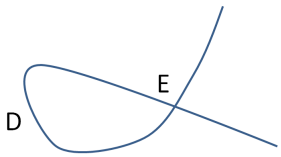


Figure 6

3.2.3 PQ must intersect CD

As shown in Figure 7, points P and Q lie on opposite sides of the straight line AB. $PC \perp AB$, $QD \perp AB$, with feet of perpendicularity C and D respectively.

If $C = D$, then PQ intersects AC at the point C.

When $C \neq D$ and PQ do not intersect segment CD, connect PD and QC. Since Q does not lie on the perpendicular PC, $\angle QCD < 90°$.

$$\angle PCQ = \angle PCD + \angle QCD < 180°.$$

Similarly, $\angle PDQ = \angle PDC + \angle QDC < 180°$.

We obtain:

$$360° - \angle PCQ > 180° \text{ and } 360° - \angle PDQ > 180°.$$

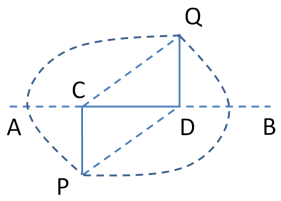


Figure 7

Since PQ does not intersect CD, it must bypass the segment CD. Whichever direction it takes to bypass, a triangle with an interior angle greater than $180°$ will be formed. This contradicts the previous conclusion. Therefore, PQ must intersect CD.

3.3 Corollary 3

According to Prop. I. 11 ([1], p. 269), as shown in Figure 8, through a point C on the straight line AB, there exists at least one perpendicular line CD, such that $\angle ACD = \angle BCD$, each being a right angle (Def. 10, [1], p. 153). Given $A * B * C$ and point E does not lie on line CD or AB, without loss of generality, assume E lies in the interior of $\angle BCD$. Thus $\angle BCE < \angle BCD$ and $\angle ACE > \angle ACD$; hence $\angle ACE \neq \angle BCD$. Therefore, line CE is not perpendicular to line AB. We have that there is one and only one perpendicular that can be drawn to a straight line from a point upon it.

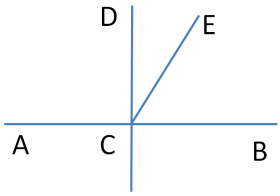


Figure 8

3.4 Corollary 4

Through a point not on a straight line, there exists one and only one straight line perpendicular to the given line.

According to Prop. I. 12 ([1], p. 271), a perpendicular line to a given straight line can be drawn from a point not on the line.

Suppose, for the sake of contradiction, that at least two straight lines, CD and CE, can be drawn from a point C not on the straight line AB, perpendicular to the given line, with D and E being the respective feet of the perpendiculars, as shown in Figure 9. Since D and E are distinct points, in $\triangle CDE$, we have $\angle CDE = \angle CED =$ right angle $= \angle CDA$. However, from Prop. I. 16 ([1], p. 276), we have $\angle CDA > \angle CED$, which yields a contradiction. Therefore, the assumption cannot hold.

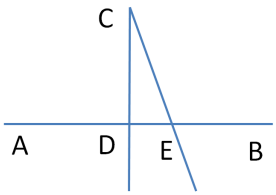


Figure 9

3.5 Corollary 5

Let the angle $\angle BAC$ formed by two rays AB and AC be $\beta°$, where $0 < \beta \leqslant 180$. Let $t_{\angle BAC}$ or t_β denote the area enclosed by $\angle BAC$, as shown in Figure 10.

Since any angle can be bisected (Prop. I. 9, [1], p. 267), a full angle of 360° can be divided equally into 2^q (where $q \in \mathbb{N}$) angles, with each angle measuring $360°/2^q$. Therefore, there must exist some q such that:

$$\frac{360}{2^q} \leqslant \beta < \frac{360}{2^{q-1}}.$$

Let this value of q be denoted by $q(\beta) = Q$, that is,

$$\frac{360}{2^Q} \leqslant \beta < \frac{360}{2^{Q-1}}.$$

Let $\frac{360}{2^Q} = \theta$. Since $\theta \leqslant \beta$, it follows that $t_\beta \geqslant t_\theta$ (C.N. 5).

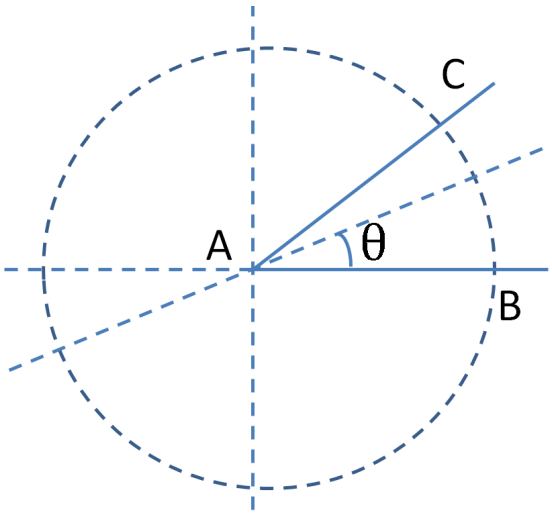


Figure 10

The full angle of 360° is equally divided into 2^Q parts, each with angular measure $\theta°$. Label these 2^Q angles as $\theta_1, \theta_2, \theta_3 \ldots, \theta_{2^Q}$, and let the area of θ_1 be $t_{\theta_1} = t$.

As $t > 0$, we have

$$t_{\theta_i} = t + \Delta t_i, \quad \frac{\Delta t_i}{t} = 0, \quad \Delta t_1 = 0.$$

Let S denote the area of the entire plane, with $S > 0$ and $t > 0$. Then we have

$$S = \sum_{i=1}^{2^Q} t_{\theta_i} = \sum_{i=1}^{2^Q} (t + \Delta t_i).$$

Let $\max(\Delta t_i)$ denote the maximum value of Δt_i, that is $\Delta t_i \leqslant \max(\Delta t_i)$. We then have

$$S \leqslant \sum_{i=1}^{2^Q} (t + \max(\Delta t_i)) = 2^Q t + 2^Q \max(\Delta t_i).$$

Since $\frac{\Delta t_i}{t} = 0$, we have $\frac{\max(\Delta t_i)}{t} = 0$,

$$\Longrightarrow \ 2^Q \frac{\max(\Delta t_i)}{t} = 0 < 1 \ \Longrightarrow \ 2^Q \max(\Delta t_i) < t.$$

We obtain

$$S < 2^Q t + t = (2^Q + 1)t.$$

This yields

$$t_\beta > t_\theta = t > \frac{S}{2^Q + 1}.$$

3.6 Corollary 6

For any β, $q(\beta) = Q$. As shown in Figure 11, select $2^{Q+1} + 4$ different points $A_1, A_2, A_3, \ldots, A_{2^{Q+1}+4}$ on a straight line such that $A_iA_{i+1} = A_jA_{j+1} = m$, $i, j < 2^{Q+1} + 4$, $i, j \in \mathbb{N}$. That is, $m = A_1A_2 = A_2A_3 = A_3A_4 = \cdots$. Erect perpendiculars A_iB_i to the straight line at each A_i; then $A_1B_1 \parallel A_jB_j$, whenever $i \neq j$ (Prop. I.27, [1], p. 282).

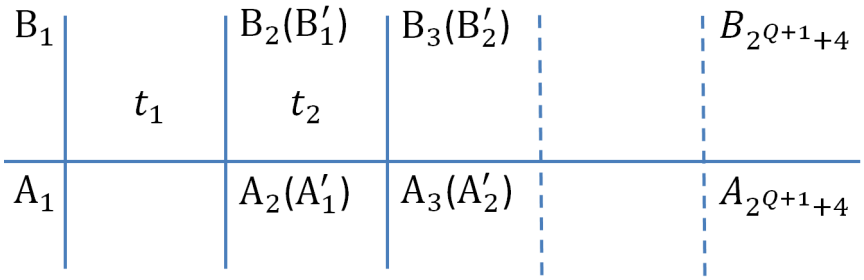


Figure 11

Shift the plane so that A_1 moves to A_1', coinciding with A_2, and A_2 moves to A_2', coinciding with A_3. There is exactly one perpendicular to a straight line passing through any given point on the line, so A_1B_1 is carried to $A_1'B_1'$, coinciding with A_2B_2, and A_2B_2 is carried to $A_2'B_2'$, coinciding with A_3B_3. By repeating this argument, the segment A_1A_2 can be made to coincide with A_iA_{i+1}, with A_1B_1 coinciding with A_iB_i and A_2B_2 coinciding with $A_{i+1}B_{i+1}$.

Let the area bounded by the straight lines A_1B_1 and A_2B_2 be denoted by $t_{A_1B_1 \| A_2B_2}$. Set $t_{A_1B_1 \| A_2B_2} = t_1 = t$. Let the area bounded by the straight lines A_iB_i and $A_{i+1}B_{i+1}$ satisfy $t_{A_iB_i \| A_{i+1}B_{i+1}} = t_i = t + \Delta t_i$, with $\Delta t_1 = 0$ (Corollary 1).

Let the area of the entire plane be S, with $S > 0$ and $t > 0$. By C.N. 5, we have

$$S > \sum_{i=1}^{2^{Q+1}+3} t_i = \sum_{i=1}^{2^{Q+1}+3} (t + \Delta t_i).$$

Let $\min(\Delta t_i)$ denote the minimum value of Δt_i, that is $\Delta t_i \geqslant \min(\Delta t_i)$. We then have

$$S > \sum_{i=1}^{2^{Q+1}+3} (t + \min(\Delta t_i))$$

$$\implies S > (2^{Q+1} + 3)t + (2^{Q+1} + 3)\min(\Delta t_i).$$

Since $\frac{\Delta t_i}{t} = 0$, we have $\frac{\min(\Delta t_i)}{t} = 0$.

$$\Longrightarrow \ -(2^{Q+1}+3)\frac{\min(\Delta t_i)}{t} = 0 < 1.$$

We obtain

$$t > -(2^{Q+1}+3)\min(\Delta t_i).$$

That yields

$$t + (2^{Q+1}+3)\min(\Delta t_i) > 0.$$

Therefore,

$$S > (2^{Q+1}+2)t.$$

If straight lines A_1B_1 and A_2B_2 are both perpendicular to a certain line, we have

$$t_{A_1B_1 \| A_2B_2} = t < \frac{S}{(2^{Q+1}+2)}.$$

Finally, we obtain the ratio of $t_{A_1B_1 \| A_2B_2}$ to t_β,

$$\frac{t_{A_1B_1 \| A_2B_2}}{t_\beta} < \frac{\frac{S}{(2^{Q+1}+2)}}{\frac{S}{2^{Q}+1}} = \frac{1}{2}.$$

4 Proof of the Fifth Postulate

It can be deduced from Props. I.11, I.12, I.27 that at least one parallel line passes through a point external to a given straight line, as shown in Figure 12.

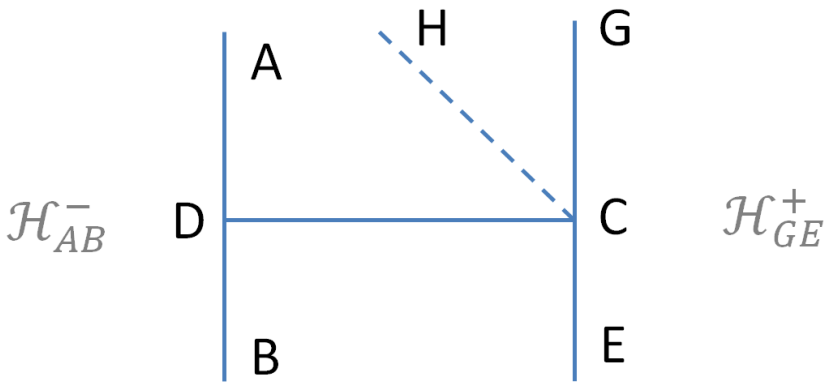


Figure 12

Let C be a point outside the straight line AB, given $C \in \mathcal{H}^+_{AB}$. Draw a perpendicular line CD from C to AB, with foot of perpendicular D (Prop. I.12, [1], p. 271). Assuming A * D * B, draw a perpendicular line GE to CD through point C (Prop. I.11, [1], p. 269). Given G * C * E, CD=m, since $\angle$ADC = $\angle$ECD = right angle, it follows that GE $\parallel$ AB (Prop. I.27, [1], p. 282).

From Corollary 4, only one line CD exists, and from Corollary 3, only one line GE exists.

Suppose there exists another straight line CH $\parallel$ AB, with $H \in \mathcal{H}^-_{GE}$. When $\angle$GCH = $\beta°$, where $0 < \beta < 90°$, and given that $\angle$GCD = 90°, the line CD intersects line AB. Thus the line AB must lie in the interior of $\angle$HCE, meaning no point on AB can lie in the interior of $\angle$GCH; otherwise AB intersects the line CH (Corollary 2), which contradicts the assumption that CH $\parallel$ AB. Therefore $t_{\angle \mathrm{GCH}}$ is part of $t_{\mathrm{AB} \parallel \mathrm{GE}}$. By C.N. 5, it follows that

$$t_{AB\parallel GE} > t_{\angle GCH} \implies \frac{t_{AB\parallel GE}}{t_{\angle GCH}} \geq 1.$$

This yields a contradiction with Corollary 6. As $AB \perp CD$ and $GE \perp CD$, by Corollary 6, it follows that

$$\frac{t_{AB\parallel GE}}{t_{\angle GCH}} < \frac{1}{2}.$$

When $\angle$GCH > 90°, we have $\angle$ECH < 90°, and the line AB lies within $\angle$GCH. By the same method, we obtain the same result.

Therefore, the assumption that there exists another parallel line cannot hold. Through a point not on a given straight line, there exists one and only one parallel line. Thus, the Fifth Postulate is proved.

5 Comparison of Three Geometries

As shown in table 1, the first three postulates of Riemannian geometry differ from those of Euclidean geometry. By contrast, Lobachevskian geometry coincides with Euclidean geometry in literal wording, yet their underlying meanings differ. Lobachevskian geometry is defined on curved surfaces; a straight line on a curved surface is a curve. Curves of equal length do not guarantee coincidence, whereas segments of equal length on a plane can be made to coincide, a consequence derived from Postulate 1. A straight line on a plane exists independently of the plane itself, but a straight line on a curved surface must reside within the surface. If the surface is deformed, its associated straight lines change accordingly. In other words, the

fundamental distinction between Euclidean and non-Euclidean geometry resides in the first postulate: an Euclidean line is uniquely determined by two points alone, while a non-Euclidean line presupposes infinitely many points that fix an underlying surface. In both the Klein disk and Poincaré disk models, a bounding circle must be specified in advance; any displacement of its center or change in its radius alters the lines represented inside. An Euclidean line, however, stays invariant as long as its two defining points are fixed. Two equal-length straight lines on a curved surface — which are, in fact, curves in space — cannot be guaranteed to coincide. This forms the core of the present proof.

Euclidean Geometry	Lobachevskian Geometry	Riemannian Geometry
Postulate 1	Literally Identical	Distinct
Postulate 2	Identical	Distinct
Postulate 3	Identical	Distinct
Postulate 4	Identical	Identical
Equal segments coincide	Distinct	Identical

Table 1. Comparison of Three Geometries

This proof method is not applicable to curved surfaces; the existence of non-Euclidean geometry does not mean the Fifth Postulate is unprovable.

References

[1] Euclid, *The Thirteen Books of the Elements*, translated by T. L. Heath, Dover Publications, New York, 1956.